

ОТЧЁТ ПО ЛАБОРАТОРНОЙ РАБОТЕ №3

Группа: Z3300

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Цель:

Познакомиться с основными алгоритмическими конструкциями, которые будут использоваться для создания CLI программы. Далее продемонстрировать эффективность использования механизма рекурсии.

1 Задача 1

Создать программу CALC с интерфейсом CLI

```
1 #include <iostream>
2 #include <cstdlib>
3 #include <cmath>
4
5 using namespace std;
6
7 double add(double a, double b) {
8     return a + b;
9 }
10
11 double subtract(double a, double b) {
12     return a + (-b);
13 }
14
15 double multiply(double a, double b) {
16     double result = 0;
17     int sign = 1;
18     if (b < 0) {
19         sign = -1;
20         b = -b;
21     }
22     for (int i = 0; i < b; ++i) {
23         result = result + a;
24     }
25     return result * sign;
26 }
27
28 double power(double base, int exp) {
29     if (exp == 0) return 1;
30     if (exp < 0) {
31         cerr << "Error: exp < 0" << endl;
32         exit(1);
33     }
34     double result = 1;
35     for (int i = 0; i < exp; ++i) {
36         result = multiply(result, base);
37     }
38     return result;
39 }
40
41 int main(int argc, char* argv[]) {
42     if (argc != 4) {
43         cerr << "Usage: " << argv[0] << " <operand1> <operand2> <operator(+,-,~)>" <<
            endl;
```

```

44         return 1;
45     }
46
47     double a = atof(argv[1]);
48     double b = atof(argv[2]);
49     string op = argv[3];
50     double result = 0;
51
52     if (op == "+") {
53         result = add(a, b);
54     } else if (op == "-") {
55         result = subtract(a, b);
56     } else if (op == "^") {
57         int exp = static_cast<int>(b);
58         if (fabs(b - exp) > 1e-9) {
59             cerr << "Error: exp no int" << endl;
60             return 1;
61         }
62         result = power(a, exp);
63     } else {
64         cerr << "Error" << endl;
65         return 1;
66     }
67
68     cout << "Result: " << result << endl;
69     return 0;
70 }

```

Скомпилируем и проверим:

```
1 g++ -o lab_03/build/calculator lab_03/src/calculator.cpp
```

Сложение:

```
1 ./lab_03/build/calculator 5 3 +
```

Вывод:

```
1 Result: 8
```

Вычитание:

```
1 ./lab_03/build/calculator 10 4 -
```

На выходе получили:

```
1 Result: 6
```

Возведение в отрицательную степень:

```
1 ./lab_03/build/calculator 10 -4 ^
```

Вывод:

```
1 Error: exp < 0
```

Нецелая степень:

```
1 ./lab_03/build/calculator 10 3.2 ^
```

```
1 Error: exp no int
```

Возведение в степень:

```
1 ./lab_03/build/calculator 10 3 ^
```

Вывод:

```
1 Result: 1000
```

2 Задание 2

Решить задачу ханойской башни использованием рекурсии

```
1  #include <iostream>
2  #include <string>
3
4  using namespace std;
5
6  struct Node {
7      string move;
8      Node* prev;
9      Node* next;
10     Node(const string& m) : move(m), prev(nullptr), next(nullptr) {}
11 };
12
13 void append(Node*& head, Node*& tail, const string& move) {
14     Node* newNode = new Node(move);
15     if (!head) {
16         head = tail = newNode;
17     } else {
18         tail->next = newNode;
19         newNode->prev = tail;
20         tail = newNode;
21     }
22 }
23
24
25 void hanoi(int n, char from, char to, char aux, Node*& head, Node*& tail) {
26     if (n == 1) {
27         string move = "Move disk 1 from " + string(1, from) + " to " + string(1, to);
28         append(head, tail, move);
29         return;
30     }
31     hanoi(n - 1, from, aux, to, head, tail);
32     string move = "Move disk " + to_string(n) + " from " + string(1, from) + " to " +
33         string(1, to);
34     append(head, tail, move);
35     hanoi(n - 1, aux, to, from, head, tail);
36 }
37
38 void printList(Node* head) {
39     Node* current = head;
40     int step = 1;
41     while (current) {
42         cout << step++ << ". " << current->move << endl;
43         current = current->next;
44     }
45 }
46
47 void freeList(Node* head) {
48     while (head) {
49         Node* tmp = head;
50         head = head->next;
51         delete tmp;
52     }
53 }
54
55 int main() {
56     const int DISKS = 5;
57     Node* head = nullptr;
58     Node* tail = nullptr;
59
60     cout << "Tower of Hanoi solution for " << DISKS << " disks:\n";
61     hanoi(DISKS, 'A', 'C', 'B', head, tail);
62     printList(head);
63 }
```

```
63     freeList(head);
64
65     return 0;
66 }
```

Пример использования для DISKS=3(чтобы не прям много в выводе было)

```
1  1. Move disk 1 from A to C
2  2. Move disk 2 from A to B
3  3. Move disk 1 from C to B
4  4. Move disk 3 from A to C
5  5. Move disk 1 from B to A
6  6. Move disk 2 from B to C
7  7. Move disk 1 from A to C
8  8. Move disk 4 from A to B
9  9. Move disk 1 from C to B
10 10. Move disk 2 from C to A
11 11. Move disk 1 from B to A
12 12. Move disk 3 from C to B
13 13. Move disk 1 from A to C
14 14. Move disk 2 from A to B
15 15. Move disk 1 from C to B
16 16. Move disk 5 from A to C
17 17. Move disk 1 from B to A
18 18. Move disk 2 from B to C
19 19. Move disk 1 from A to C
20 20. Move disk 3 from B to A
21 21. Move disk 1 from C to B
22 22. Move disk 2 from C to A
23 23. Move disk 1 from B to A
24 24. Move disk 4 from B to C
25 25. Move disk 1 from A to C
26 26. Move disk 2 from A to B
27 27. Move disk 1 from C to B
28 28. Move disk 3 from A to C
29 29. Move disk 1 from B to A
30 30. Move disk 2 from B to C
31 31. Move disk 1 from A to C
```

Отчет по лабораторной работе размещен в lab_03/doc/

```
1 cp /mnt/c/Users/Maria/Downloads/lab3.pdf lab_03/doc/
```